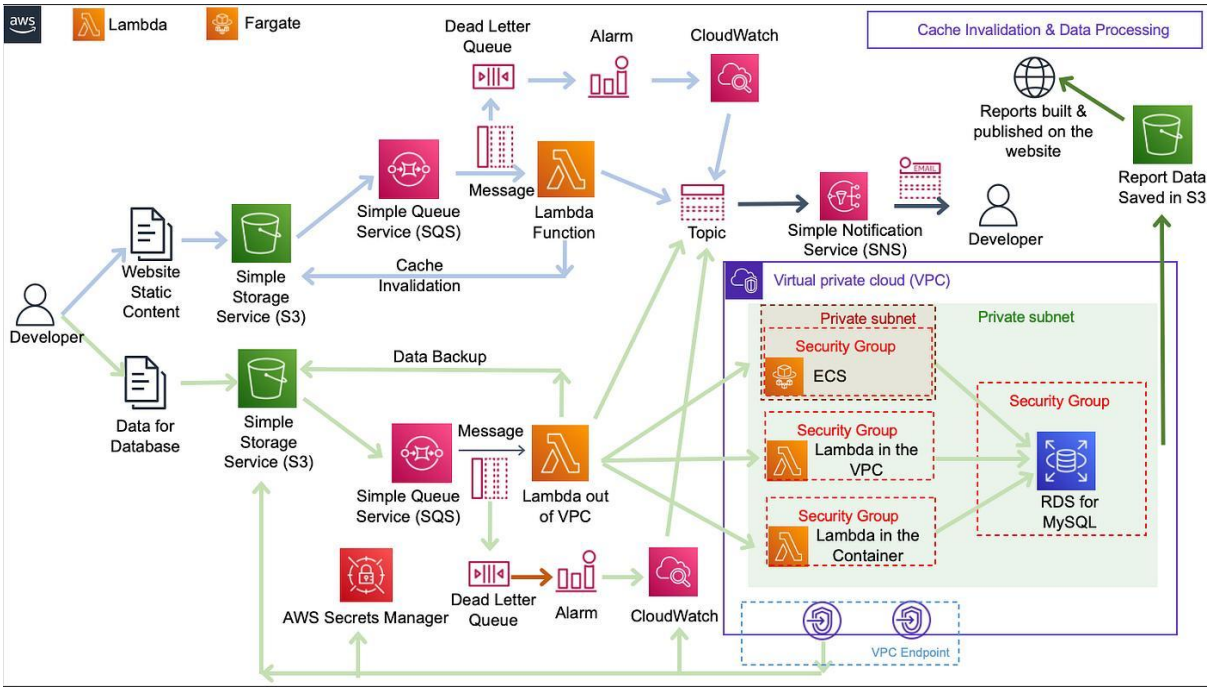
Project: AWS File Monitoring & Alert System



1. Project overview

Imagine a company stores employee documents, invoices, reports, and backup files in an Amazon S3 bucket.

The company wants to:

- Monitor file uploads and deletions.
- Send email alerts when important events happen.
- Store events in a queue for processing.
- Monitor errors and generate alarms.
- Track who accessed or modified AWS resources.
- Distribute a website or downloadable content securely through CloudFront.

2. AWS services and their purpose

1. Amazon S3 — File storage

Store company documents, images, reports, and backups.

Example: An employee uploads invoice.pdf.

2. Amazon SQS — Message queue

Receives event messages so they can be processed reliably and asynchronously.

Example: A message records that invoice.pdf was uploaded.

3. Amazon SNS — Notifications

Sends notifications to subscribed email addresses or other endpoints.

Example: An administrator receives an upload alert.

4. Amazon CloudWatch — Monitoring

Monitors metrics, logs, and alarms.

Example: Generate an alarm if an application reports repeated failures.

5. AWS CloudTrail — Audit history

Records AWS API activity to help identify which identity performed an action and when.

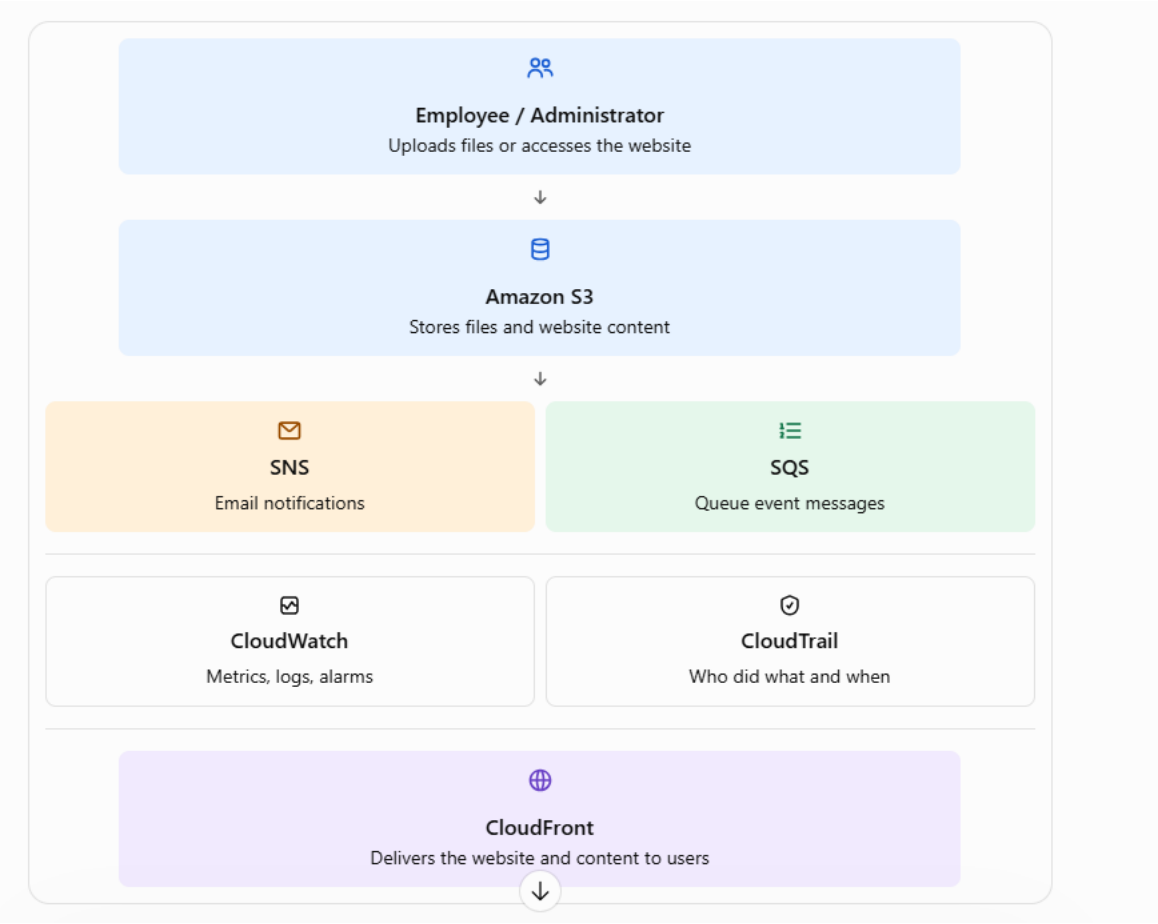
Example: Investigate who deleted an S3 object.

6. Amazon CloudFront — Content delivery

Delivers a website or public-facing content through edge locations, improving delivery speed.

Example: Employees access a company portal hosted on S3 through CloudFront.

3. Architecture: how the project works



Important: This is a conceptual architecture. S3 event notifications can publish to SQS or SNS, but not directly to both destinations for the same notification configuration. To fan out one event to both, use SNS with separate subscriptions to SQS and email. CloudFront is a separate content-delivery path, while CloudTrail and CloudWatch provide auditing and monitoring.

4. Practical implementation plan

Step	Practical task	Expected result
1	Create an S3 bucket	Files can be uploaded
2	Create an SNS topic and confirm email subscription	Email notifications work
3	Create an SQS queue	Messages can be received
4	Connect S3 events to SNS, then SNS to SQS and email	Upload events reach both destinations
5	Configure CloudTrail and its S3 destination	AWS API activity is recorded
6	Use CloudWatch metrics, logs and alarms	Monitoring and alerting work
7	Create a CloudFront distribution for the S3 website/content	Website or content is delivered through CloudFront
8	Test upload, deletion, alerts, and audit history	End-to-end verification

Conclusion

The AWS File Monitoring and Alert System demonstrates how multiple AWS services work together to build a secure, monitored, and reliable cloud solution.

- Amazon S3: Stores files and website content.
- Amazon SNS: Sends email notifications about important events.
- Amazon SQS: Stores event messages for reliable, asynchronous processing.
- Amazon CloudWatch: Monitors metrics, logs, and alarms to identify issues.
- AWS CloudTrail: Tracks AWS API activities for auditing and security investigations.
- Amazon CloudFront: Delivers website content efficiently to users around the world.

Final outcome

When a user uploads or deletes a file, the system can generate notifications, queue event messages, monitor operational issues, and maintain an audit trail. CloudFront separately delivers the website or content to users.

This project provides practical experience with AWS monitoring, messaging, security auditing, content delivery, and cloud operations. It is a useful portfolio project for aspiring Cloud Operations Engineers, AWS Support Engineers, and DevOps Engineers.

END